

Project Evidentia: Backend Setup

This document details the MySQL database initialization, directory structuring, and secure PDO connection implementation for **Project Evidentia**.

1. Target Directory Topology

The backend directory structure divides system configuration parameters from actionable endpoint route handlers:

```
backend/  
├── config/  
│   ├── db.php # CORS, Preflight, and PDO Connection Manager  
│   └── api/  
└── index.php # Core API Routing Matrix / Health Check Endpoints
```

2. MySQL Database Initialization

Execute the following sequences within your MySQL terminal instance to build the structured data workspace:

```
# Access the local MySQL engine  
mysql -u root -p  
  
# Create the dedicated UTF-8 optimized database instance  
CREATE DATABASE evidentia_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;  
  
# Verify configuration creation and exit  
SHOW DATABASES;  
EXIT;
```

3. Connection Module implementation (`config/db.php`)

This script manages Cross-Origin Resource Sharing (CORS), processes pre-flight options dispatched via platforms like Cloudflare, ngrok, or Vite, and registers the database driver instance safely:

```
<?php  
// /home/kcplibrary/Documents/evidentia/backend/config/db.php  
  
// Global Access Permission controls for cross-origin requests  
header("Access-Control-Allow-Origin: *");
```

```

header("Access-Control-Allow-Methods: POST, GET, OPTIONS, DELETE");
header("Access-Control-Allow-Headers: Content-Type, Authorization, X-Requested-With, ng");
header("Content-Type: application/json");

// Intercept browser options pre-flight validation rules safely
if ($_SERVER['REQUEST_METHOD'] == 'OPTIONS') {
    http_response_code(200);
    exit();
}

$host = "localhost";
$dbname = "evidentia_db";
$username = "root";
$password = "amNekku2026";

try {
    // Secure PDO instantiation with parameterized attributes
    $pdo = new PDO("mysql:host=$host;dbname=$dbname;charset=utf8mb4", $username, $password);

    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    $pdo->setAttribute(PDO::ATTR_DEFAULT_FETCH_MODE, PDO::FETCH_ASSOC);
    $pdo->setAttribute(PDO::ATTR_EMULATE_PREPARES, false);

} catch (PDOException $e) {
    // Raise explicit HTTP status rules on failure states
    http_response_code(500);
    echo json_encode([
        "status" => "error",
        "message" => "Asena, kabaw: Database connection failed",
        "debug" => $e->getMessage()
    ]);
    exit();
}
?>

```

4. Route Test Module Setup (`api/index.php`)

To verify structural configuration links and execution flow across folders, establish this testing gateway:

```

<?php
// /home/kcplibrary/Documents/evidentia/backend/api/index.php

// Bind the core configuration layer
require_once __DIR__ . '/../config/db.php';

// Generate verified system status responses

```

```
echo json_encode([
    "status" => "success",
    "message" => "Welcome to Evidentia API Router! Connection verified.",
    "database" => "evidentia_db"
]);
?>
```

5. Active Testing Verification

1. Boot your localized PHP application system directly via the `api/` workspace location:

```
cd /home/kcplibrary/Documents/evidentia/backend/api
php -S localhost:8000
```

2. Point your network browser or API tracking utilities to: `http://localhost:8000`